College Licensing and Reputation Effects on the Labor Market

Preliminary draft
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Abstract

Can signals about college quality improve the hiring process of recent graduates, where employers lack information about individual worker productivity? We study whether a mandatory college licensing process affects college reputations and the labor market outcomes of their graduates, and how these impacts vary with the level of ex-ante uncertainty. We obtain monthly earnings data at the individual level for Peruvian college graduates from 2014 to 2019 and study a major licensing process. Using a staggered difference-in-differences approach, we show that a positive licensing outcome increases the unconditional wages of the college's recent graduates by 3%. Consistent with models of employer learning and employers observing productivity over time, we find that workers with a longer tenure (at least one year at their job prior to treatment) did not benefit significantly; on the other hand, workers with a short tenure (0-3 months) experienced a large and significant increase in wages (18% of the baseline) driven by a 2.7 percentage point increase in employment. Results for graduates whose college was denied accreditation are noisier, with patterns suggesting a negative impact.

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1 Introduction

College reputation plays a critical role in the hiring process of recent graduates where employers lack information about individual worker productivity (MacLeod et al., 2017), making institutional reputation particularly important in labor markets. Yet, while enrollment in higher education has expanded dramatically, capacity at the most reputable institutions has remained relatively constrained. In the United States, total college enrollment doubled over the past half-century, with enrollment at elite colleges barely increasing (Blair and Smetters, 2021). This pattern is not unique to developed countries. In Latin America, the gross enrollment rate more than doubled between 2000 and 2018 (World Bank, 2020), driven primarily by private sector expansion, which by 2015 comprised over half of all enrollment (Program for Research on Private Higher Education, 2015). The proliferation of new institutions, many of dubious quality, has made it harder for employers to distinguish between high- and low-productivity graduates, potentially creating inefficiencies in labor market matching. Quality assurance policies, such as accreditation systems, can reduce this information asymmetry by providing credible signals about institutional quality, thereby improving hiring decisions and labor market efficiency.

This paper examines how information disclosure about college quality affects labor market outcomes. Specifically, we analyze how a regulatory policy that releases college quality information influenced firm hiring decisions and graduates' employment outcomes in Peru. In 2015, the Peruvian Ministry of Education implemented a comprehensive higher education reform that enforced compliance with a set of basic quality standards.[1] Under this policy, all universities were required to meet minimum quality requirements to receive an operational license and continue their activities. Licensing

[1] Similar regulatory interventions have been adopted across Latin America, including in Brazil and Ecuador, in response to stagnating achievement levels and lagging educational outcomes (Ferreyra et al., 2017).

decisions were announced publicly between 2016 and 2021, with universities denied a license required to cease operations within two years. The policy resulted in the closure of 50 out of 144 universities.

We match college records with monthly employment data to track labor market outcomes from 2014 to 2019 for all Peruvians who graduated from college in 2014 and 2015, immediately before the reform began. Our empirical strategy addresses two key identification challenges. First, separating quality signals from student selection is difficult because better colleges typically attract higher-ability students. By focusing exclusively on graduates who completed their education before the reform, we hold human capital constant and isolate the pure signaling effect of licensing decisions on labor market outcomes. Second, information about college quality is usually released through rankings of all institutions, and the lack of a natural control group complicates causal inference. The staggered timing of licensing announcements between 2016 and 2021 enables us to implement a difference-in-differences strategy, comparing graduates from institutions that received licensing decisions at different points in time. This approach allows us to credibly identify the causal effect of quality information disclosure on graduate employment and wages.

We find that new information about college quality has a significant impact on labor market outcomes. Positive signals increase graduate wages, while license denials produce symmetrical negative effects, though neither significantly impacts employment rates. Economic models of employer learning predict that signals about college quality matter the most when the uncertainty about worker's productivity is higher. Aligned with the theoretical predictions, these effects are concentrated among workers where employer uncertainty is highest: short-tenured graduates experience gains in both wages and employment following positive signals, while graduates with over one year of tenure show no significant response. Our findings show that outcomes react to the policy in a manner consistent with the presence of asymmetric information about the quality of education among workers and with a process of on-the-job information production.

Beyond wages and employment, we document that licensing announcements affect the quality and type of jobs held by recent graduates. Short-tenure workers from licensed colleges are significantly more likely to work in large firms and to transition from private sector to public sector employment. These patterns suggest that the information released through licensing improves job matching, enabling graduates to access higher-quality employment opportunities that better align with their credentials. Increased mobility of short-tenure workers following positive licensing announcements is consistent with improved bargaining power and the ability to secure better working conditions.

Our paper contributes to two strands of literature on information frictions in the labor markets. First, our work connects to literature on worker signals and employer learning (Abel et al., 2020; Bartik and Nelson, 2025; Bassi and Nansamba, 2022; Bates, 2020; Bordón and Braga, 2020; Carranza et al., 2022; MacLeod et al., 2017). In particular and closely related to labor markets of college graduates, MacLeod et al. (2017) argue that college reputation partially substitutes for individual signals when the latter are unavailable, and show that returns to reputation become smaller when more precise individual signals emerge through on-the-job observation[2]. In this sense, most of this literature has primarily studied how individual-level information acquisition reduces the relevance of group-level signals.[3] We extend this framework by examining a college-level information shock rather than individual certification or learning processes. In our

[2]Arteaga (2018) also documents that demand for college graduates in an elite college in Colombia decreased following cuts to the university curriculum, showing that employers also react to changes in human capital of workers.

[3]In other settings, Bassi and Nansamba (2022) shows that certifying workers' skills in Uganda led to more assortative matching and higher earnings conditional on employment, while Carranza et al. (2022) find that credible public signals about individual worker skills improve labor market outcomes. Bates (2020) demonstrates that the availability of objective benchmarks on the quality of potential hires increases mobility of the most productive workers in the labor market of teachers.

setting, the regulator produces public signals about the quality of workers' education, which serves as a proxy for productivity (Rivera, 2011 shows that educational credentials are the most common criterion used to evaluate resumes). Our findings highlight that significant improvements in market outcomes can be produced by assessment of college quality, not just individual worker skills. Moreover, consistent with models of signaling and employer learning, we find that licensing decisions have the strongest effects on workers with high uncertainty—those early in their careers where employers have limited individual performance data. For experienced workers with established track records, the college-level signal becomes redundant, and licensing decisions have minimal impact on outcomes. In contrast to papers that take reputation as fixed and use variation in the availability of individual-level signals, our research examines the impact of a policy that directly influences a college's reputation and, consequently, the labor market outcomes of its graduates.[4]

Second, we build on research documenting the returns to college prestige in labor market outcomes (Anelli, 2020; Black et al., 2023; Eble and Hu, 2022; Jia and Li, 2021; Sekhri, 2020). Sekhri (2020) and Anelli (2020) demonstrate that graduates from more prestigious institutions experience superior labor market outcomes even when controlling for observable measures of human capital, suggesting that employers face imperfect information about workers' true skills and rely on institutional reputation as a signal.[5] Eble and Hu (2022) show how schools that changed to more prestigious-sounding names attracted higher-aptitude students and generated wage premiums in audit studies, demonstrating that signals about college prestige can impact labor market

[4]In a similar spirit, Mosquera and Miranda (2025) study university reputation as a labor-market signal by exploiting changes in Ecuador's governmental university rankings, finding that reputational declines significantly harm early-career employment outcomes—effects that fade as workers reveal their true productivity over time.

[5]Note that we refer to elite institutions only. However, several studies have estimated labor market returns of college attendance in different contexts (see recent examples in Grosz, 2020; Montoya et al., 2018; Zimmerman, 2014).

outcomes. While this literature has established that college prestige matters, it has primarily focused on wage premiums associated with attending elite institutions in equilibrium. Our contribution is to examine how labor market outcomes respond when new information about college quality becomes publicly available to all employers. By studying the Peruvian licensing policy that released credible signals about institutional quality, we provide direct evidence on how information disclosure affects the signaling value of college credentials and demonstrate that employers actively update their beliefs in response to quality information.

The next section discusses the relevant institutional context in the Peruvian higher education system. Section 3 describes the data used in our analysis. Section 4 presents the empirical strategy, while Section 5 presents and discusses our results. Section 6 concludes.

2 Background

2.1 Higher Education Regulation and Licensing in Peru

In 1996, with Law No. 882, the Peruvian Congress allowed the creation of for-profit universities.[6] In the following decade, Peru experienced sustained economic growth and increasing access to higher education, with the number of colleges expanding rapidly to meet the demand. However, concerns regarding the labor market outcomes of college graduates began to arise among policymakers and the public. This issue is described in a large body of literature that documents disparities in the quality of higher education options in Peru (Díaz, 2008; Yamada et al., 2015).

In an effort to address concerns about the quality of colleges, the Peruvian Congress

[6] In the early 1990s, following a severe economic crisis, Peru transitioned to an open market, adopting market deregulation policies and promoting the private sector. These changes were in line with similar reforms enacted in other Latin American countries during the same time period.

introduced a series of reforms. In 2012, a moratorium was imposed on the creation of new universities, with the aim of restricting the entry of low-quality institutions. In 2014, Peru's Congress created the National Superintendence of Higher University Education (SUNEDU) to verify colleges' compliance with minimum quality standards through a comprehensive licensing process.[7]

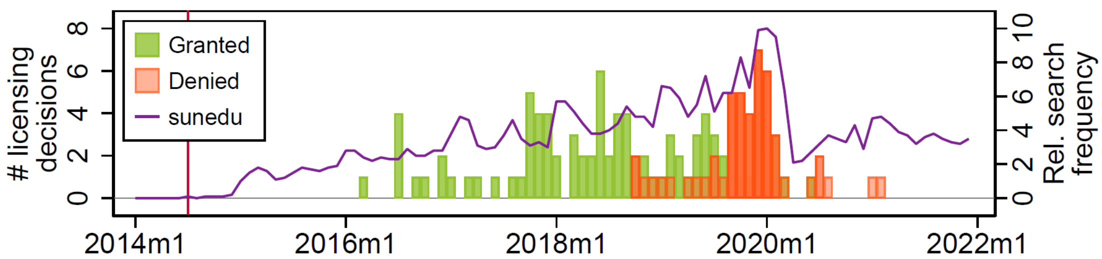
The conditions to obtain an operating license required compliance with eight specific requirements: (1) having clear academic objectives and (2) academic plans for each degree; (3) having good standing infrastructure and equipment; (4) having a research agenda; (5) faculty with at least a Master's or Ph.D. degree, and at least 25% of faculty should be full-time professors; (6) provision of complementary services (such as health and psychological services); (7) establishment of job placement mechanisms, such as job banks; and (8) ensuring institutional transparency.[8]

Universities were required to meet all requirements without exception. Failing to meet even one of these standards could lead to license denial. The first licensing decisions following an examination of compliance were released in July 2016. For the first two years following the start of this process, all examined colleges were considered to be in compliance with the required standards and were granted an operating license. In late 2018, the first license denial was issued by SUNEDU. By the end of the licensing process in early 2021, a total of 50 universities out of 144 were denied an operating license, which forced them to cease all operations within two years. The timing of licensing decisions and their outcome are displayed in Figure 1.

[7] SUNEDU, created with Law No. 30220 of 2014, also has the authority to impose sanctions on institutions that fail to adhere to these rules and its responsibilities extend to managing requests for changes to college names, monitoring the minimum requirements for degree-granting, publishing annual reports on the use of university benefits and the state of higher education in Peru, and releasing statistics related to the sector.

[8] A document that compiles the minimum set of standards that universities should comply with is publicly available at `https://www.sunedu.gob.pe/8-condiciones-basicas-de-calidad/`. See Appendix A.2 for more details about these requirements.

Figure 1: Licensing Decisions and Google search volume over Time



Notes: Authors' elaboration based on Google Trends data on the relative frequency of searches for the keyword "sunedu." Licensing decisions to grant or deny an operating license and their timing are obtained from SUNEDU.

The licensing process received significant attention in Peruvian media. Google Trends data suggest that the process was highly salient in Peru, as seen in Figure 1. Increases in searches for the keyword "sunedu" started after the institution's creation, and search volume mirrors the frequency of licensing decisions released, indicating substantial public awareness and interest in the licensing outcomes.

Despite ample scrutiny in the process, the outcome of SUNEDU's examination was difficult to predict for most institutions. Not surprisingly, a small group of elite universities was granted licenses in the early stages of the process. However, the outcomes for most other institutions remained uncertain until their announcements. In Appendix A.3, we assess whether licensing decisions were predictable using machine learning techniques and data from the 2010 University Census (CENAUN), which provides comprehensive information on approximately 100 variables related to infrastructure, faculty, administration, and student characteristics for universities operating prior to the reform.[9] While the model is very effective at identifying elite institutions highly likely to receive licenses, we find limited predictive power for outcomes at non-elite universities, making models less reliable for predicting denials.

We also implement survival analysis using Cox-Time regression to predict the timing

[9] We explore several prediction methods, with the k-nearest neighbors algorithm achieving the highest accuracy (test-set score of 0.92).

of licensing decisions. While the correlation between predicted and actual announcement timing is positive, the model is considerably more accurate for early decisions—which largely correspond to elite universities—than for later announcements. This suggest that while outcomes for top-tier institutions were relatively predictable, licensing decisions for the majority of universities remained uncertain until their official announcements.[10]

The current state of the licensing process remains ambiguous as political opposition in Congress attempts to overturn this reform. In this paper, we focus on the licensing decisions announced between July 2016 and November 2019, due to data constraints (see Section 3 for more details). SUNEDU enjoyed substantial support during the studied period, as exemplified by its ability to deny a license to one-third of Peruvian colleges and enforce their closure.

2.2 The Labor Market of College Graduates

The discussed higher education expansion democratized access but also generated a surplus of graduates in "popular" fields that were disconnected from labor market needs (Castro and Yamada, 2013). Consequently, "overeducation" and "underemployment" became frequent issues. Research from the period, such as that by Yamada (2015), highlighted that this boom, combined with weak job markets and low-quality university programs, increased the risk of graduates finding themselves in jobs that did not require their university credentials, leading to lower-than-expected wages and high rates of professional dissatisfaction.

In markets with such information asymmetry, employers often use educational cre-

[10] It is worth highlighting that the timing of licensing decisions and announcements is not orthogonal to college quality metrics. In Appendix Figure A.1, we observe that universities whose graduates enjoyed higher average wages and wage growth were more likely to be granted a license in the early months of the process. After an initial period of high wage growth, most wage trajectories become approximately linear.

dentials as a “screening” device to sort workers by unobserved ability, rather than as a direct measure of human capital (Psacharopoulos and Patrinos, 2018). In Peru, this meant that the prestige and quality of the attended university became a crucial signal for employers. In a correspondence experiment, Agüero et al. (2023) found that employers greatly value credible signals of ability: when fictitious resumes included a ”Beca 18” scholarship (a merit-based award for talented, low-income students), the likelihood of a callback increased by 20%.

3 Data

In partnership with the Peruvian Ministry of Education, we matched educational records of recent graduates with their labor market outcomes using a combination of three administrative datasets. First, we obtain information on educational attainment for all students who graduated from college between 2014 and 2019. This dataset includes the university attended, major, graduation year, and degree level (e.g., BA, MA). We restrict our attention to bachelor-level graduates from the classes of 2014 and 2015, ensuring that all students in our sample graduated prior to the start of the licensing reform. This approach allows us to isolate the effects of licensing announcements on labor market outcomes, excluding confounding factors related to the policy's influence on educational choices or college operations during students' enrollment.

Second, we matched this sample of college graduates to monthly tax records, obtained from the Peruvian Ministry of Labor, and built a panel dataset spanning January 2014 to November 2019. This data includes wages, hours worked, anonymized employer identification, and sector information. Each student is observed monthly for approximately six years, providing rich longitudinal data on early career trajectories. It is important to note that this data does not include information about the informal sector, which employs approximately half of Peruvian college graduates.[11] Our findings are therefore subject to this significant limitation, as the informal sector is a likely destination for recent graduates rather than unemployment. However, wages in the informal sector are substantially lower on average and typically fall below the minimum wage, suggesting that movements into informality may reflect negative labor market outcomes.

Third, for each college observed in our panel of college graduates, we added the outcome of the licensing process and the timing of its announcement obtained from SUNEDU's website. As depicted in Figure 1, 94 colleges were granted a license, while 50 colleges denied it and ultimately shut down. For our empirical analysis, we focus exclusively on licensing announcements that occurred between July 2016 and November 2019, as constraints from the earnings data limit our ability to study shocks occurring after November 2019.

4 Empirical Strategy

Our empirical strategy exploits the staggered rollout of licensing decisions in Peru's higher education reform to estimate the causal impact of quality information disclosure on graduate labor market outcomes. We employ a staggered difference-in-differences (DiD) approach that compares graduates from treated colleges to those from not-yet-treated colleges, before and after the licensing announcement. The staggered nature of the licensing announcements, which occurred between 2016 and 2021, provides variation in the timing of information revelation across institutions. This setting allows us to use graduates from colleges that have not yet received their licensing decision as a comparison group for those whose college quality has just been revealed to the labor

[11] To address this issue, we complement our main analysis using survey data. See Section 5.3 for its results.

market. Reflecting the possible outcomes of the licensing process, we define two types of treatment: *Granted* for universities that were granted a license and *Denied* for those that were denied it.

Incorporating heterogeneity and dynamics in the typical difference-in-differences staggered design necessitates avoiding standard "two-way fixed effects" (TWFE) regressions, as these have been shown to be problematic in such setups (Baker et al., 2021; Goodman-Bacon, 2021). We follow an approach similar to the one introduced by Cengiz et al. (2019).[12] This approach looks at each treatment occurrence in isolation, sidestepping the problem introduced by staggered treatment. Because this approach builds event-specific datasets and stacks them on top of each other, it is also known as "stacked regression." For each licensing event (which can comprise several universities if their events are concurrent), we select a 12-month window before and after the treatment event; control units are chosen among those that will receive the same licensing outcome between 13 and 24 months later. Each dataset can then be indexed by g, the timing of the licensing event of the included treated units.

We estimate the following model:

$$y_{i,t,c(i),g(c(i))} = \beta D\{t - g(c(i)) \geq 0\} + \psi_{t,c(i),g(c(i))} + \mu_{i,c(i),g(c(i))} + e_{i,t,c(i),g(c(i))} \quad (1)$$

where i indexes individuals, t indexes time (months), and $c(i)$ denotes the college from which individual i graduated. The treatment indicator $D\{t - g(c(i)) \geq 0\}$ equals one for periods after the licensing announcement for college $c(i)$. The parameter of interest, β, captures the average treatment effect on the treated (ATT) in the 12 months following the licensing announcement. The specification includes individual fixed effects $\mu_{i,c(i),g(c(i))}$, which absorb time-invariant unobserved heterogeneity across graduates, and

[12] As a robustness check, we also follow Callaway and Sant'Anna (2021) to estimate the main results of the license decision, and find similar results.

time fixed effects $\psi_{t,c(i),g(c(i))}$ estimated separately for each event-specific dataset ($g(c(i))$ or "stack"). These time fixed effects flexibly control for aggregate trends that may differ across treatment cohorts. The treatment dummy is the only parameter shared across "stacks": the estimated ATT will be a weighted average of the single ATTs that could be obtained from each dataset. Because the stacked regression approach creates separate datasets for each licensing event, we can flexibly estimate how treatment effects vary across different groups while maintaining clean identification. Standard errors are clustered at the university level within each stack to account for potential correlation in outcomes among graduates from the same institution and serial correlation over time.

Our primary outcomes capture different dimensions of labor market performance. We examine monthly unconditional wages, measured as total labor income in Nuevos Soles (PEN $\approx$ USD 0.3), as well as a binary indicator for formal employment. We also analyze hours worked (total monthly hours in formal employment), public sector employment (a binary indicator for employment in the public sector), and large firm employment (a binary indicator for employment in firms with more than 100 workers). These outcomes enable us to evaluate whether information disclosure influences not only the level of compensation but also employment opportunities and job characteristics that may reflect employer perceptions of graduate quality.

Our difference-in-differences methodology requires two important assumptions: parallel trends and no anticipation. To assess the plausibility of the first assumption, we examine pre-treatment trends in all outcome variables through event study specifications. These graphs, presented in the results section, generally show no evidence of differential pre-trends, supporting the validity of our approach. We also assume that neither graduates nor employers anticipate the timing or outcome of the licensing announcement. This assumption is plausible given that the timing and outcomes of individual licensing decisions were not publicly known in advance, and the reform represented a novel policy intervention in Peru's higher education sector.

4.1 Sample Definition

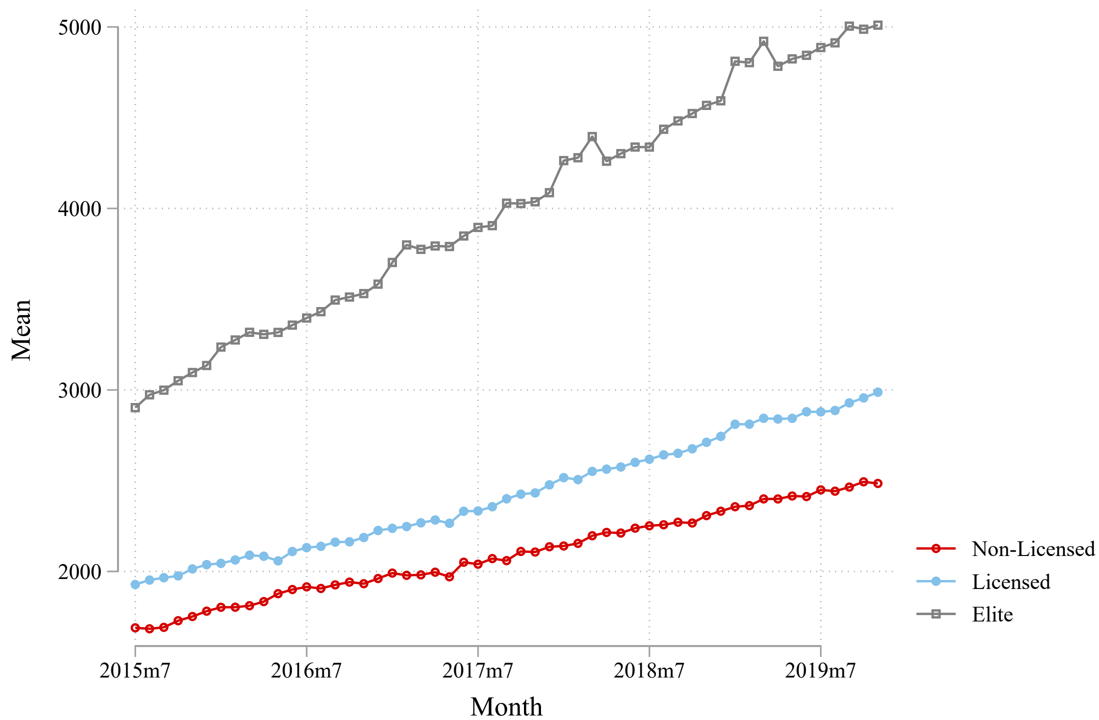
Because universities that will later be granted a license are unlikely to offer a proper comparison to universities that are denied one, and vice versa, we estimate separate models depending on the type of treatment. In both cases, we include individuals who graduated with a bachelor's degree in the 2014 and 2015 classes as described in the previous section.

Licensing decisions were announced between 2016 and 2021. For our empirical analysis, we only include licensing announcements until the end of 2019 to ensure sufficient post-treatment observation periods. After applying our sample restrictions, including the exclusion of elite universities discussed below, our analysis includes 47 licensed colleges and 50 colleges that were denied licenses and subsequently closed.

We also exclude 10 universities from our main analysis sample. As shown in Figure 2, elite college graduates have substantially higher levels of income with wage trajectories that grow at a faster pace—earning approximately 50 percent more than other college graduates, a difference that persists over time. Additionally, elite colleges were significantly more likely to receive licenses early in the process. Our machine learning analysis using 2010 University Census (CENAUN) data confirms that elite colleges ranked highest in both the probability of obtaining a license and receiving it early.[13]

[13] See Appendix A.3 for more details about these results.

Figure 2: Average Monthly Wages of Recent Graduates



These institutions represent a distinct segment of Peru's higher education market. Moreover, these elite institutions are unlikely to be affected by the licensing process in the same way as non-elite colleges, as their reputations were already well-established in the labor market prior to the reform. The differential outcome trends for elite college graduates would violate the parallel trends assumption, making them unsuitable as either treatment or control units in our main specification. In contrast, non-elite colleges are considerably harder to predict in terms of quality and are surprisingly more comparable to one another, making them the appropriate focus for our analysis of information revelation effects.

Table 1 presents descriptive statistics for the three groups in the year prior to the policy start. Graduates from licensed colleges and those from denied colleges have remarkably similar characteristics: employment rates are nearly identical (36.6% vs. 36.4%), as are average monthly wages (PEN 1,713 vs. PEN 1,716, where PEN $\approx$ USD 0.3), hours worked (195.5 vs. 196.6), public sector employment rates (7.8% for both), and large firm employment (70.8% vs. 68.7%). Both groups differ markedly from elite college graduates, who have higher employment rates (41.8%) and substantially higher wages (PEN 2,478). The similarity in pre-treatment characteristics between the granted and denied groups supports the plausibility of using not-yet-treated colleges as controls in the stacked regression framework, while the differences with elite graduates justify their exclusion. An important compositional difference is that 53% of licensed colleges are public universities, compared to 28.8% of elite colleges and 0% of denied colleges (which were all private institutions).

Table 1: Baseline Summary Statistics

	Granted	Elite	Denied
Employed	0.366	0.418	0.364
	(0.482)	(0.493)	(0.481)
Income	1712.7	2478.2	1716.3
	(1221.4)	(1707.9)	(1300.6)
Hours Worked	195.5	199.1	196.6
	(61.89)	(60.25)	(60.74)
Public Employer	0.0777	0.0363	0.0780
	(0.268)	(0.187)	(0.268)
Large Firm	0.708	0.666	0.687
	(0.455)	(0.472)	(0.464)
Public University	0.530	0.288	0
	(0.499)	(0.453)	(0)
Observations	1,346,484	235,728	224,760

Notes. Data sources are confidential and anonymized administrative data from the Peruvian Ministry of Education and the Peruvian Ministry of Labor. Mean in the year prior to policy start; SD in parentheses. Wages are measured in Nuevos Soles ($PEN = 0.3USD$).

Figure A.1 illustrates the average monthly wage trajectories of graduates from different college types over the study period. The employment and wage income of licensed graduates are similar to those of graduates from colleges that were denied licenses and subsequently closed. Both groups follow parallel trajectories and are substantially different from elite college graduates, who have much higher employment rates and wage income throughout the observation period. Notably, all public universities in our study were eventually granted a license.

4.2 Heterogeneous Effects by Job Tenure

A key advantage of our monthly panel data structure is the ability to examine how the information effect varies with employer learning. Because the timing of treatment is unique in each dataset g, we can impose conditions on job tenure before treatment occurs. This allows us to obtain treatment effects for individuals who were just hired by comparing them to individuals who were hired at the same time but who graduated from different colleges. We can then estimate our results conditional on tenure being short (0-3 months) or long (more than 12 months) and observe whether the certification value of the licensing decision varies along this dimension.

Specifically, we estimate equation (1) separately for:

- **Short tenure:** Workers employed 0-3 months at their current job when the licensing decision is announced

- **Long tenure:** Workers employed more than 12 months at their current job when the licensing decision is announced

For short-tenure workers, employers have had limited time to learn about individual productivity, so we expect the college quality signal to be most valuable. In contrast, for long-tenure workers, employers have accumulated substantial information about individual performance, potentially diminishing the importance of the college reputation signal. This heterogeneity analysis provides direct evidence on the mechanism through which information disclosure affects labor market outcomes and tests whether employer learning about individual worker quality substitutes for statistical discrimination based on college reputation.

5 Results

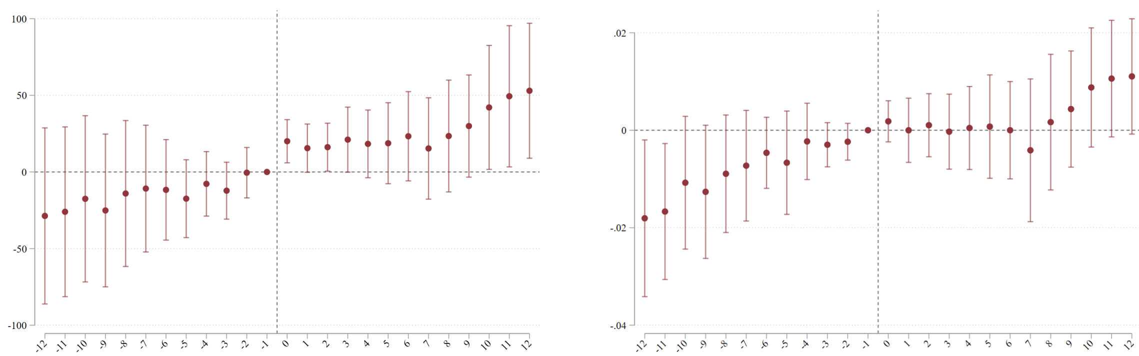
This section presents the impact of college licensing announcements on graduates' labor market outcomes. We first examine the aggregate effects of license approvals and denials on labor market outcomes using event study analysis. We then explore heterogeneous impacts across different groups and by job tenure, examining whether the effects differ between recent labor market entrants and more established workers. Finally, we discuss the broader implications of these findings.

5.1 License Approval Effects

Figure 3 presents event study results for graduates from colleges that received licensing approval. Panel (a) shows a clear positive impact on monthly wages following the licensing announcement. Graduates experience a discrete jump in wages after their college receives a license, with an estimated increase of 29 PEN (approximately USD 9) that is statistically significant at the 5% level. This effect is persistent throughout the post-treatment period, suggesting that the quality signal provided by licensing has lasting effects on labor market outcomes.

In contrast, Panel (b) of Figure 3 demonstrates that overall employment rates do not show a significant increase following licensing approval. The employment probability remains relatively stable across the pre- and post-treatment periods, with confidence intervals encompassing zero. This suggests that while licensing improves wage outcomes for those already employed, it does not substantially increase the likelihood of formal employment in the aggregate.

Figure 3: Event Study Results: License Granted



(a) Wages (PEN)

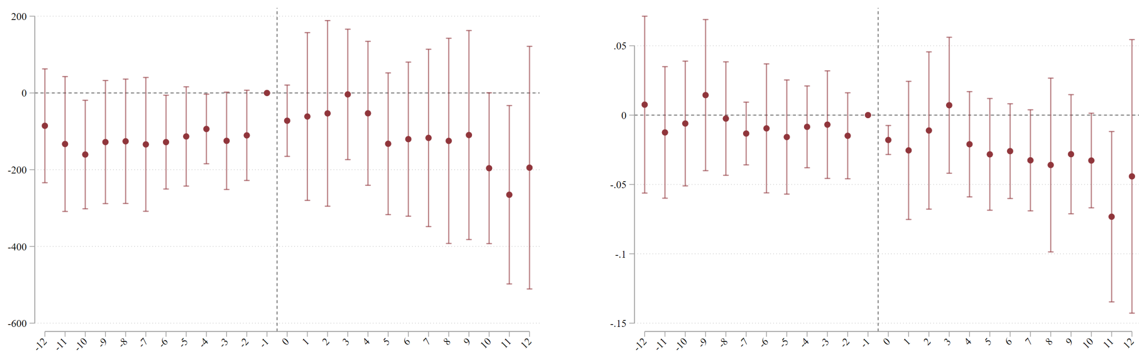
(b) Employment

Notes. Regressions include individual and time fixed effects.

5.2 License Denial Effects

Figure 4 presents corresponding results for graduates from colleges that were denied licensing. These results are notably noisier due to a smaller sample size, reflecting the fact that fewer colleges were denied licenses during the study period. Despite the increased uncertainty, the estimates suggest symmetric negative effects to those observed for license approvals. Both wages (Panel a) and employment (Panel b) show suggestive decreases following the announcement of license denial, though these effects are estimated with less precision than the positive effects of license approval.

Figure 4: Event Study Results: License Denied



(a) Wages (PEN) (b) Employment

Notes. Regressions include individual and time fixed effects.

5.3 Informal Employment and Self-Employment

To investigate whether licensing affects employment in the informal sector or self-employment, we analyze data from the ENAHO household survey, which covers both formal and informal workers. Unlike our primary administrative data from tax records, the ENAHO survey can capture workers outside the formal sector.

The ENAHO results show no consistent pattern in informal employment or self-employment following either approval or denial of a license. Although the nature of this data necessarily implies reduced statistical power, this may suggest that the main labor market adjustments occur within the formal sector rather than through transitions between formal and informal work. Other outcomes measured in ENAHO (wages and employment) are considerably noisier than in our administrative data but remain consistent with our main findings, providing a robustness check on our results. See Appendix A.4 for details on results and estimation.

5.4 Heterogeneous Impact by Tenure

A key question this study seeks to answer is whether labor market impacts of licensing vary substantially depending on workers' job tenure at the time of the announcement.[14] Models of asymmetric information suggest that workers with a longer tenure, who are well known to their employer, should not be affected by the release of information generated by the licensing process. On the other hand, workers with a short tenure should experience the largest impacts from the new information. Table 2 presents difference-in-differences estimates of the average treatment effect on the treated (ATT) for the first 12 months following licensing announcements, disaggregated by tenure groups.

Table 2: Difference-in-Differences Post-treatment Estimates

	All		Short tenure		Long Tenure	
	Wages	Empl.	Wages	Empl.	Wages	Empl.
ATT - License Granted	28.9**	0.003	61.1**	0.027***	25.2	-0.003
	(13.27)	(0.004)	(24.96)	(0.009)	(24.6)	(0.0073)
Placebo	-15.6	-0.008**	5.7	0.005	13.2	-
	(18.08)	(0.004)	(9.62)	(0.005)	(14.89)	(-)
Baseline Mean	902	0.455	344	0.193	1892	0.888
Impact (% baseline)	3%	1%	18%	14%	1%	0%
N	8,717,176	8,717,176	5,203,505	5,203,505	714,626	326,700

Notes. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Regressions include individual and time fixed effects. ATTs are calculated for the first 12 months after the license announcement. Monthly wages are measured in Nuevos Soles (PEN) ($PEN \approx 0.3USD$).

[14] We also investigate heterogeneity in the effects of licensing announcements across institution types and fields of study, as well as additional labor market dimensions. Examining heterogeneity in the effects of license approval, we find substantial variation across institution types and fields of study. Graduates from private universities experience large positive impacts from licensing approval, while the effects for public university graduates are close to zero. This may reflect differences in the information environment, with private institutions facing greater uncertainty about their quality prior to licensing. Across fields of study, graduates in Education, the Arts, and Health majors benefit most from licensing approval. This may reflect variation in the importance of formal credentials across professions, with teaching and healthcare positions placing particular weight on institutional accreditation. Results are available upon request.

Across all workers, licensing approval generates a statistically significant wage increase of 28.9 PEN (approximately USD 9), representing a 3% increase relative to the baseline mean of 902 PEN. The effect on employment probability is small and not statistically significant at 0.3 percentage points. Placebo tests examining the period before treatment show no significant pre-trends for wages, with a coefficient of -15.6 PEN that is not statistically significant, supporting the parallel trends assumption underlying our identification strategy.

The effects are substantially larger and more significant for graduates with short tenure (0-3 months at their current job). This group experiences a wage increase of 61.1 PEN (approximately USD 18), which is statistically significant at the 5% level and represents an 18% increase relative to their baseline mean wage of 344 PEN. This effect is more than double the overall sample average, highlighting the particular importance of licensing information for workers at the early stages of their careers. Employment effects are also concentrated among short-tenure workers, with a 2.7 percentage point increase that is statistically significant at the 1% level. Given a baseline employment rate of 19.3% for this group, this represents a substantial 14% relative increase in employment probability. Placebo estimates for short-tenure workers show no significant pre-trends, with small and statistically insignificant coefficients for both wages (5.7 PEN) and employment (0.005).

These larger effects for short-tenure workers are consistent with asymmetric information playing a more significant role early in workers' careers. When workers have limited job tenure, employers have less direct information about their productivity and must rely more heavily on observable signals such as college quality. The licensing announcement provides a credible, public signal that updates employers' beliefs about graduate quality, leading to improved job matching and wage offers.

In contrast, graduates with long tenure (13-24 months at their current job) show substantially smaller and statistically insignificant effects. The estimated wage increase is 25.2 PEN (1% of baseline) with a standard error of 24.6 PEN, and the employment effect is essentially zero at -0.003 percentage points. The placebo estimate for wages is 13.2 PEN but cannot be precisely estimated.

The minimal impact on long-tenure workers suggests that by the time workers have been with an employer for over a year, the employer has accumulated sufficient direct information about worker productivity that the public signal of college quality becomes less relevant. This is consistent with models of statistical discrimination where employers learn about worker quality over time through on-the-job performance.

5.4.1 Other Market Outcomes for License Approval

Finally, we examine how licensing approval affects specific dimensions of employment quality and sector of work, again disaggregating by job tenure.

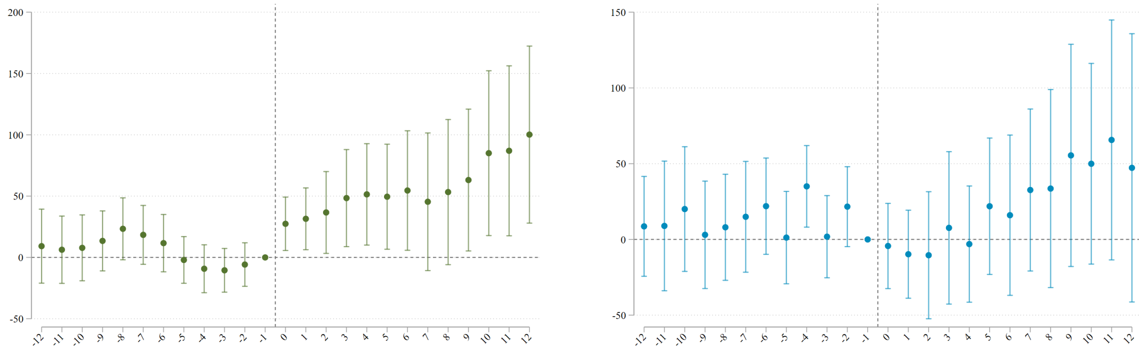
Wages. Figures 5 presents event study estimates of wage effects separately for short-tenure (Panel a) and long-tenure (Panel b) workers. The short-tenure group shows a clear positive jump in wages following licensing approval, with the effect emerging immediately after treatment and persisting throughout the follow-up period. The estimated effect of 61 PEN (approximately USD 18) is highly statistically significant at the 1% level. For long-tenure workers, the wage trajectory remains relatively flat across the pre- and post-treatment periods, with wide confidence intervals that include zero throughout. This confirms that the aggregate wage effects documented earlier are driven entirely by workers at the beginning of their careers rather than those with established employment relationships.

Employment. Figure 6 shows employment probability patterns by tenure. Short-tenure workers (Panel a) experience a significant 2.7 percentage point increase (statistically significant at the 1% level) in employment probability following licensing approval. This effect appears immediately after treatment and remains stable across the post-treatment period. Long-tenure workers (Panel b) show no discernible change in employment probability, with the event study estimates fluctuating around zero throughout both the pre- and post-treatment periods.

Firm Size. The type of employer also responds to licensing information, particularly for short-tenure workers. Figure 7 demonstrates that licensing approval increases employment in large firms by 3 percentage points (statistically significant at the 1% level) for short-tenure graduates. This suggests that large firms, which may have more sophisticated human resource departments and greater capacity to process quality signals, are particularly responsive to the information provided by licensing announcements. Long-tenure workers show no change in their probability of working at large firms.

Public vs. Private Sector. Licensing approval appears to facilitate transitions from private to public sector employment, particularly for short-tenure workers. Figure 8 shows that private sector employment decreases significantly for short-tenure workers following licensing approval. Conversely, Figure 9 demonstrates a corresponding significant increase in public sector employment for this group. Long-tenure workers show no significant changes in either sector, though there is suggestive evidence of a small decline in private employment. This sectoral reallocation suggests that public sector employers may place greater weight on formal credentials and quality signals when making hiring decisions. The licensing approval may make short-tenure graduates more attractive candidates for public sector positions, which often have more formalized hiring processes that emphasize educational qualifications.

Figure 5: Heterogeneous Impacts on Wages of License Approval



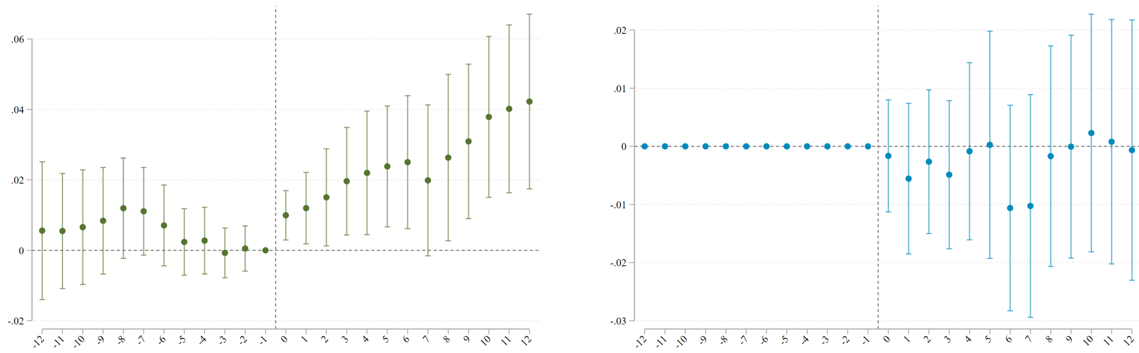
(a) Short (0-3 months)

(b) Long (13-24 months)

Notes.

Employment Effects. Figure 6 shows employment probability patterns by tenure. Short-tenure workers (Panel a) experience a significant 2.7 percentage point increase (statistically significant at the 1% level) in employment probability following licensing approval. This effect appears immediately after treatment and remains stable across the post-treatment period. Long-tenure workers (Panel b) show no discernible change in employment probability, with the event study estimates fluctuating around zero throughout both the pre- and post-treatment periods.

Figure 6: Heterogeneous Impacts on Employment of License Approval

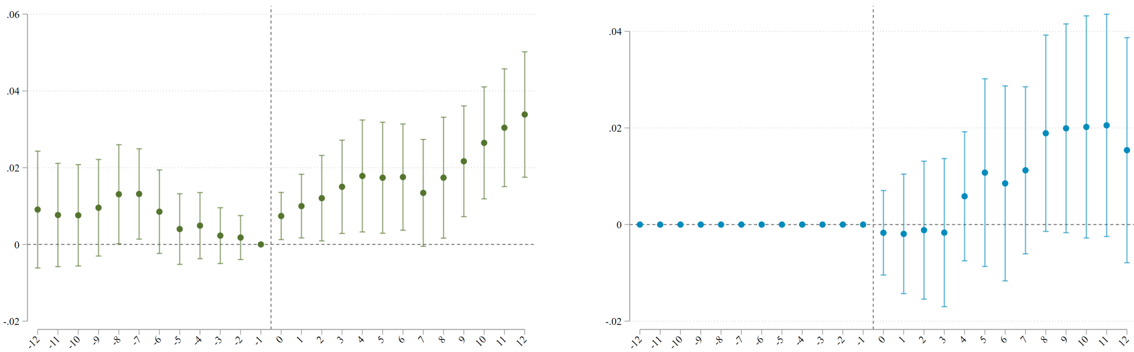


(a) Short (0-3 months)　　(b) Long (13-24 months)

Notes. Employment increased with a similar pattern to wages for short-tenured graduates (+2.7*** p.p.). No change for long-tenured graduates.

Firm Size. The type of employer also responds to licensing information, particularly for short-tenure workers. Figure 7 demonstrates that licensing approval increases employment in large firms by 3 percentage points (statistically significant at the 1% level) for short-tenure graduates. This suggests that large firms, which may have more sophisticated human resource departments and greater capacity to process quality signals, are particularly responsive to the information provided by licensing announcements. Long-tenure workers show no change in their probability of working at large firms.

Figure 7: Heterogeneous Impacts on Large Firm Employment of License Approval

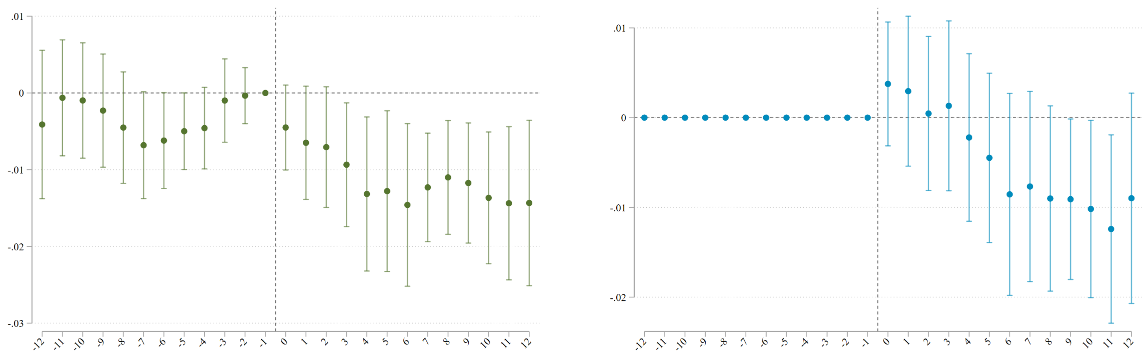


(a) Short (0-3 months) (b) Long (13-24 months)

Notes. Employment in large firms increased with a similar pattern to wages for short-tenured graduates (+3*** p.p.). No change for long-tenured graduates.

Public vs. Private Sector Employment. Licensing approval appears to facilitate transitions from private to public sector employment, particularly for short-tenure workers. Figure 8 shows that private sector employment decreases significantly for short-tenure workers following licensing approval. Conversely, Figure 9 demonstrates a corresponding significant increase in public sector employment for this group. Long-tenure workers show no significant changes in either sector, though there is suggestive evidence of a small decline in private employment. This sectoral reallocation suggests that public sector employers may place greater weight on formal credentials and quality signals when making hiring decisions. The licensing approval may make short-tenure graduates more attractive candidates for public sector positions, which often have more formalized hiring processes that emphasize educational qualifications.

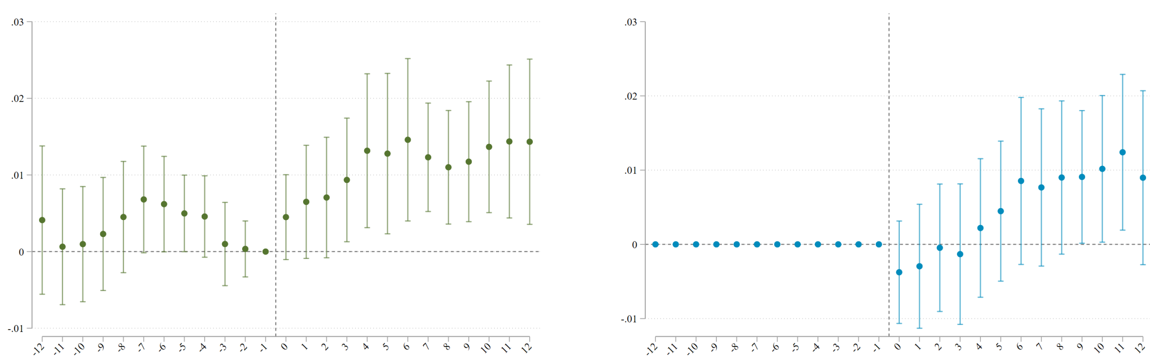
Figure 8: Heterogeneous Impacts on Private Sector Employment of License Approval



(a) Short (0-3 months) (b) Long (13-24 months)

Notes.).

Figure 9: Heterogeneous Impacts on Public Sector Employment of License Approval



(a) Short (0-3 months) (b) Long (13-24 months)

Notes.

5.4.2 Other Market Outcomes for License Denial

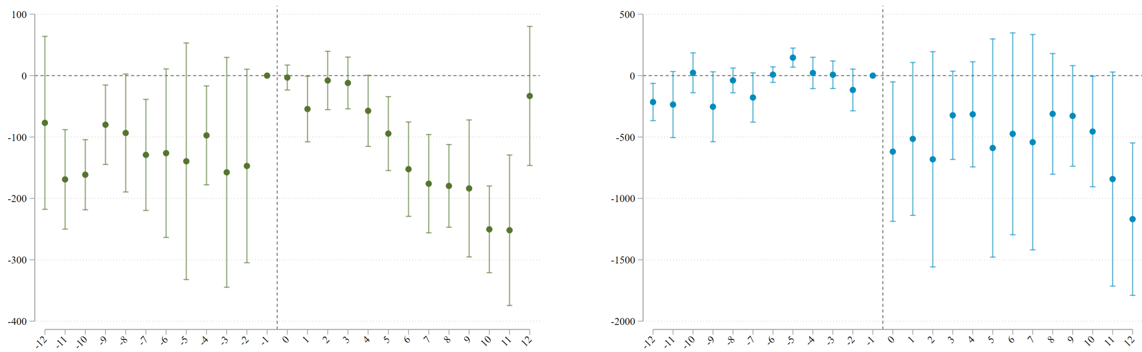
We now turn to examining how license denial affects labor market outcomes across the tenure distribution. As noted earlier, these estimates are less precise due to smaller sample sizes, but they provide important evidence on the symmetry of licensing effects.

Wage and Employment Impacts. Figure 10 presents wage effects of license denial by tenure. While the results are noisier when the sample is split, they suggest negative wage impacts for both short-tenure (Panel a) and long-tenure (Panel b) groups. The point estimates indicate wage decreases, though the confidence intervals are wider than in the license approval analysis. Figure 11 shows that employment decreases following license denial, with the pattern for short-tenure graduates being clearer than for long-tenure workers. This mirrors the asymmetry observed in the license approval results, where short-tenure workers are more responsive to quality signals.

Firm Size. The probability of working in a large firm (Figure 12) decreases following license denial, particularly for short-tenure graduates. This suggests that large firms respond to negative quality signals just as they respond to positive ones, reducing their hiring of graduates from denied colleges. The effects for long-tenure workers are less clear, consistent with the pattern that established employment relationships are less sensitive to public quality signals.

Public vs. Private Sector. License denial appears to push short-tenure workers toward private sector employment and away from public sector positions. Figure 13 shows an increase in private sector employment probability, while Figure 14 demonstrates a decrease in public sector employment for short-tenure workers following license denial. These effects are opposite to those observed for license approval, suggesting symmetric responses across sectors. Long-tenure workers again show less clear changes in sectoral employment.

Figure 10: Heterogeneous Impacts on Wages of License Denial

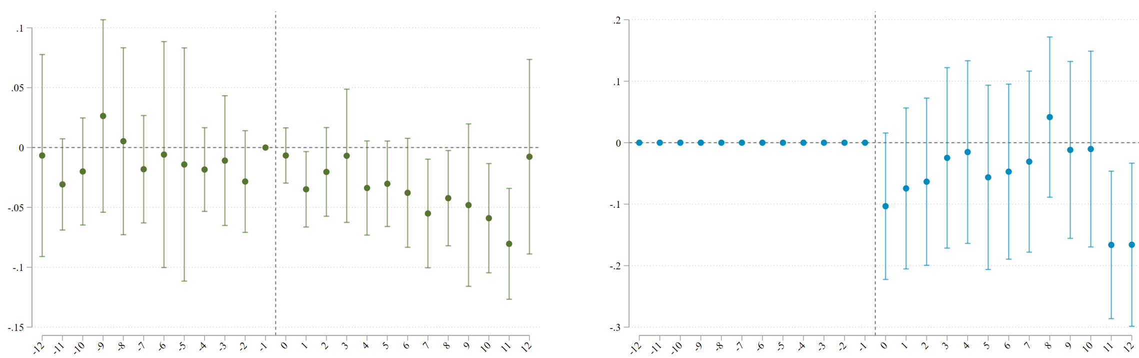


(a) Short (0-3 months)

(b) Long (13-24 months)

Notes.

Figure 11: Heterogeneous Impacts on Employment of License Denial

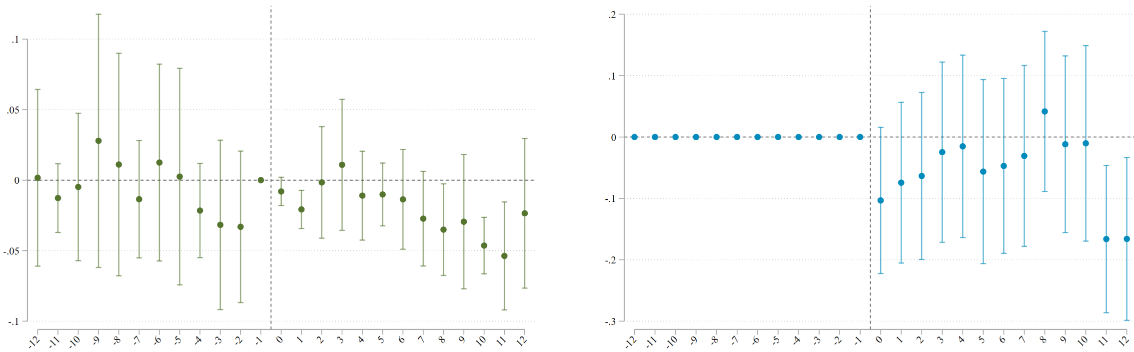


(a) Short (0-3 months)

(b) Long (13-24 months)

Notes.

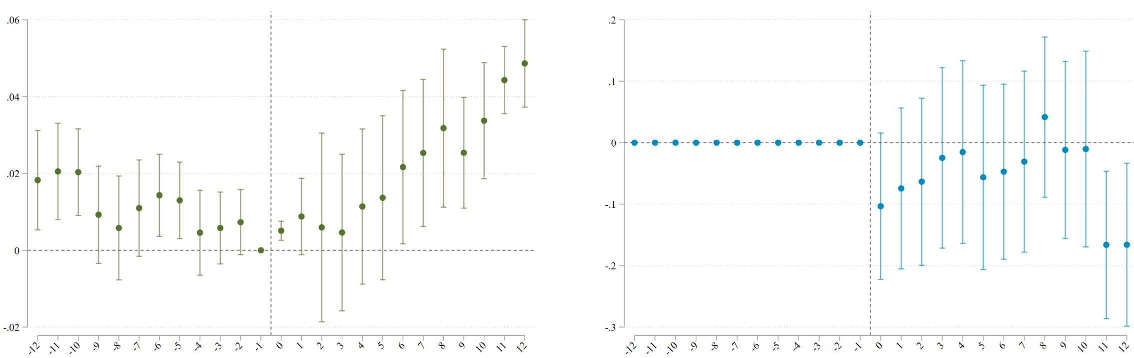
Figure 12: Heterogeneous Impacts on Large Firm Employment of License Denial



(a) Short (0-3 months) (b) Long (13-24 months)

Notes.

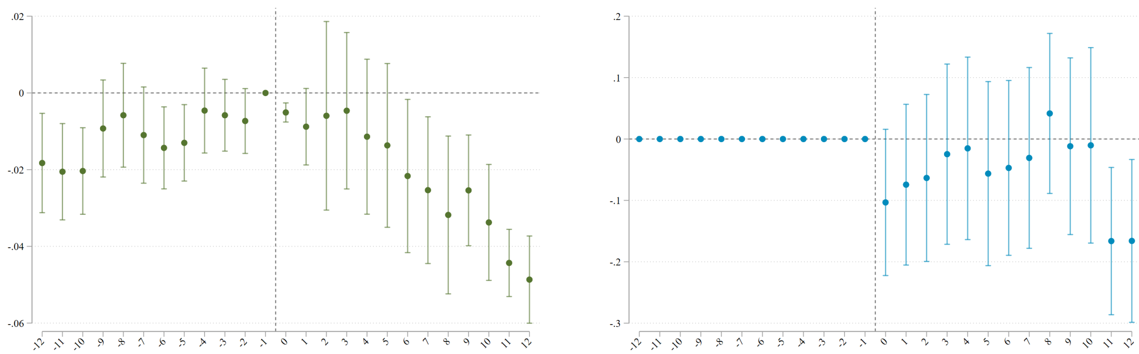
Figure 13: Heterogeneous Impacts on Private Sector Employment of License Denial



(a) Short (0-3 months) (b) Long (13-24 months)

Notes.

Figure 14: Heterogeneous Impacts on Public Sector Employment of License Denial



(a) Short (0-3 months) (b) Long (13-24 months)

Notes.

6 Conclusion

In this paper, we studied the effects that signals produced by a college accreditation process carried out in Peru between 2016 and 2021 had on college graduates' labor market outcomes. Our analysis leverages the staggered release of decisions, which resulted in operational licenses being granted to some institutions while others were denied and subsequently shut down. By combining administrative educational records with monthly formal employment data from tax records, we can track how the public release of quality information about colleges affects the wages, employment, and job characteristics of recent graduates.

Our results indicate that the announcement of a college complying with quality standards leads to a positive and statistically significant effect on wages. However, this aggregate effect masks substantial heterogeneity across workers. The benefits are concentrated among individuals with short job tenure (0-3 months), who experienced an 18% wage increase (approximately USD 18 per month) and a 14% increase in employment probability (2.7 percentage points). In contrast, we find no significant effects for workers with long tenure (13-24 months), suggesting that the positive signal from licensing is only relevant when employers have limited information about their employees' productivity. We also show that licensing announcements affect other dimensions of jobs held by recent graduates: short-tenure workers become significantly more likely to work in large firms and to transition from private sector to public sector employment after their college is granted a license.

These findings contribute to our understanding of how information disclosure policies affect labor market outcomes in the presence of asymmetric information. The concentration of effects among short-tenure workers is consistent with models of employer learning, where direct observation of worker productivity over time reduces the importance of educational credentials and institutional reputation.

While we find no evidence in household survey data that licensing affects informal employment or self-employment, our statistical power only allows for precise estimation of impacts within the formal sector. This represents a limitation of our analysis in a country with substantial labor market informality. However, formal employment typically has higher productivity and offers greater protections and benefits, representing an outcome of major interest to the overall economy and policymakers.

Our findings have important implications for education policy and labor market regulation, particularly in developing countries with rapidly expanding higher education sectors and concerns about institutional quality. The results demonstrate that public certification of college quality can generate substantial benefits for graduates by reducing information asymmetries between workers and employers. When colleges of dubious quality are widespread in weakly regulated markets, as was the case in Peru prior to the licensing reform, employers face costly uncertainty about the skills and training of job applicants. The provision of credible quality signals through a rigorous licensing process helps resolve this uncertainty, facilitating more efficient matching between workers and firms.

While we cannot precisely estimate the full impact of college closures, the suggestive evidence of negative effects on graduates highlights the potential costs of quality enforcement. Policymakers implementing similar reforms must weigh the benefits of improved information and quality standards against the potential harm to students already enrolled in or graduated from low-quality institutions. Transition policies, such as teach-out agreements or transfer opportunities, may help mitigate these costs. More broadly, our findings contribute to ongoing debates about the role of regulation in higher education and the mechanisms through which institutional reputation and quality signals affect economic outcomes. As countries around the world grapple with expanding access to higher education while maintaining quality standards, the evidence from Peru's licensing reform offers valuable lessons about the potential benefits and limitations of quality assurance policies in shaping labor market opportunities for college graduates.

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A Appendix

A.1 Additional Figures and Tables

A.1.1 Additional Figures

Figure A.1: Average Monthly Wages by University

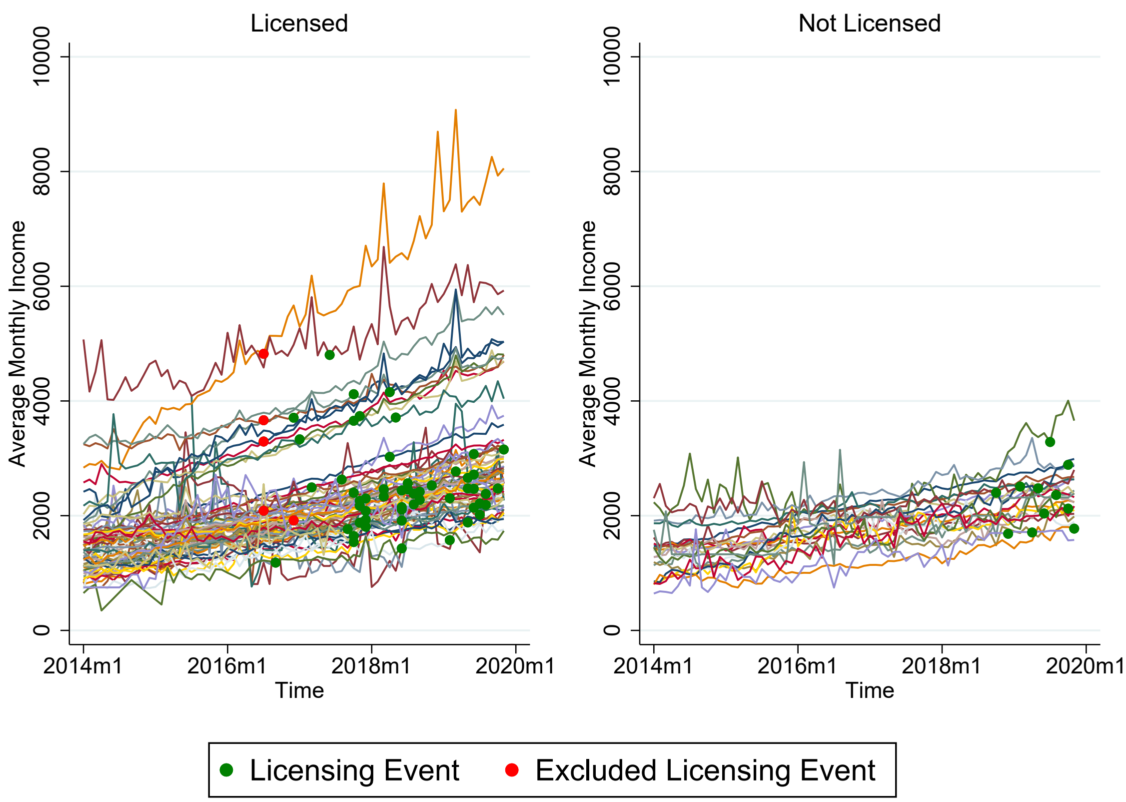


Table A.1: ATT Estimates of Granted License Colleges following Callaway and Sant'Anna (2021)

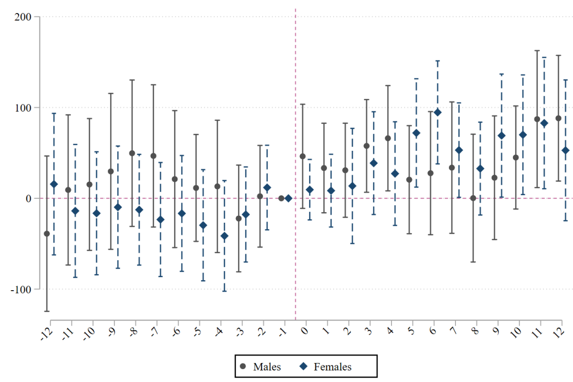
	Post-Treatment		**Pre-Treatment**	
	ATT	(s.e.)	Placebo	(s.e.)
Wages (PEN)	67.360 ***	(12.62)	-2.358	(21.66)
Employment	0.004 **	(0.00)	0.000	(0.00)
Hours Worked	-0.738	(0.87)	-0.809	(1.08)
Tenure	-0.463 *	(0.20)	-0.135 *	(0.08)
Public Employer	0.001	(0.00)	0.001	(0.00)
Large Firm	-0.003	(0.00)	-0.002	(0.00)

Notes. *** $p < 0.01$, ** $p < 0.05$,* $p < 0.10$. Regressions include individual and month fixed effects. ATTs are calculated using the Callaway and Sant'Anna (2021) estimator for the first 12 months after the license announcement. ATT (Pre) are the average treatment effects 12 months before the treatment. SE (Pre) are the standard errors for the ATT(Pre) estimates. Monthly wages are measured in Nuevos Soles (PEN) ($PEN \approx 0.3USD$). Tenure is measured as the total amount of working months in their main job.

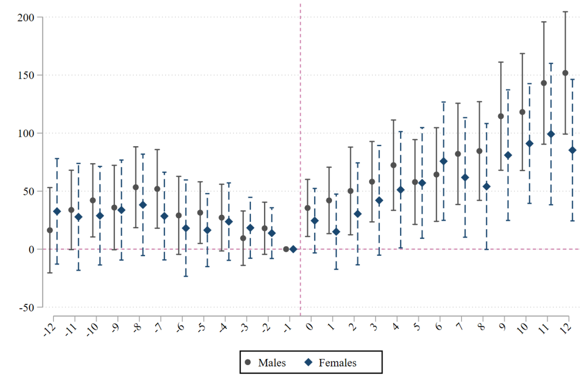
Table A.2: ATT Estimates of Denied License Colleges following Callaway and Sant'Anna (2021)

Notes. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Regressions include individual and month fixed effects. ATTs are calculated using the Callaway and Sant'Anna (2021) estimator for the first 12 months after the license announcement. ATT (Pre) are the average treatment effects 12 months before the treatment. Standard errors are in parenthesis. Monthly wages are measured in Nuevos Soles (PEN) (1 $PEN \approx 0.3USD$). Tenure is measured as the total amount of working months in their main job.

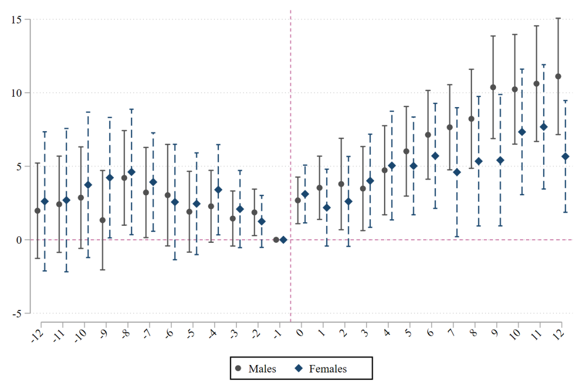
Figure A.2: Event Study for Graduates of Licensed Universities By Gender



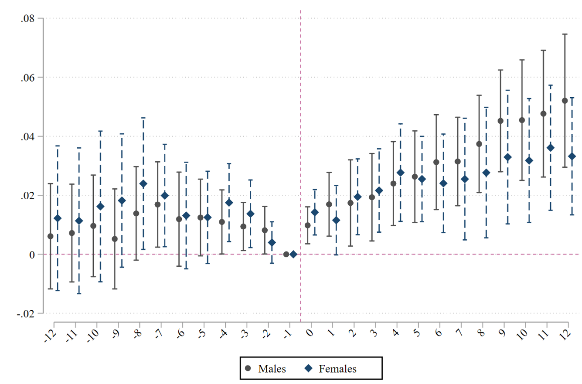
(a) Wages



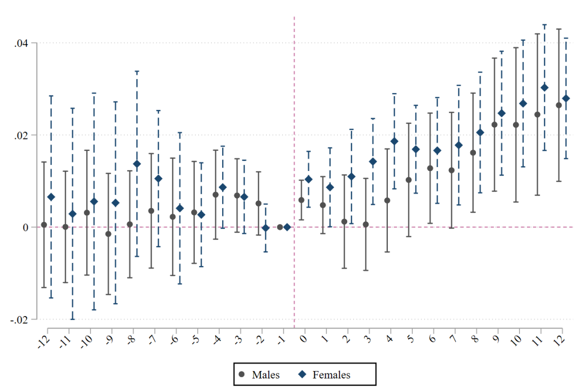
(b) Wages (Unconditional)



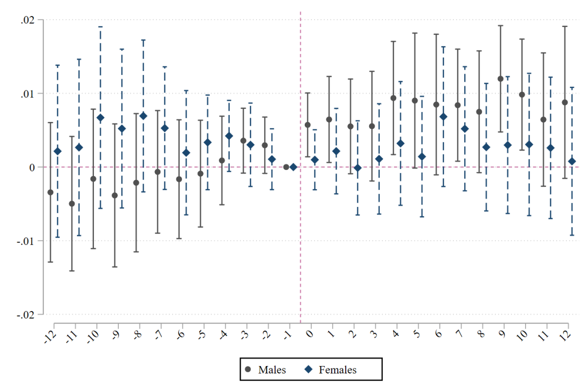
(c) Total Hours



(d) Employment

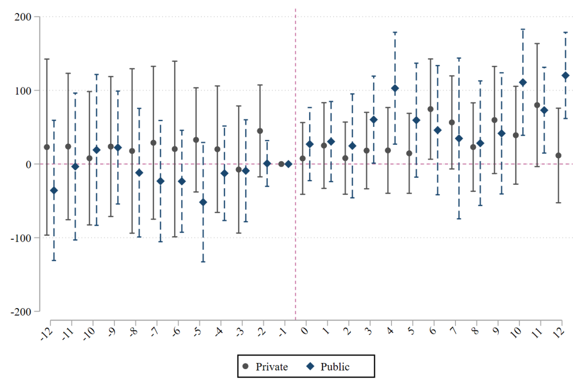


(e) Large Firm

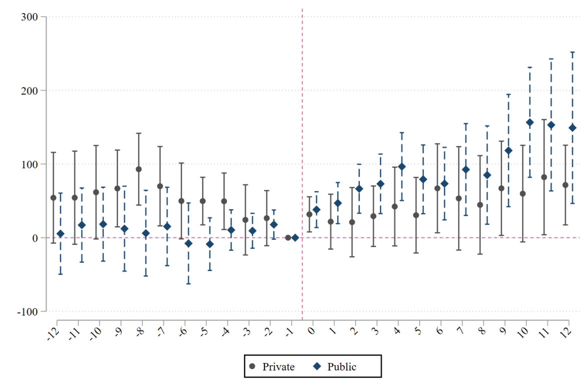


(f) Public Employer

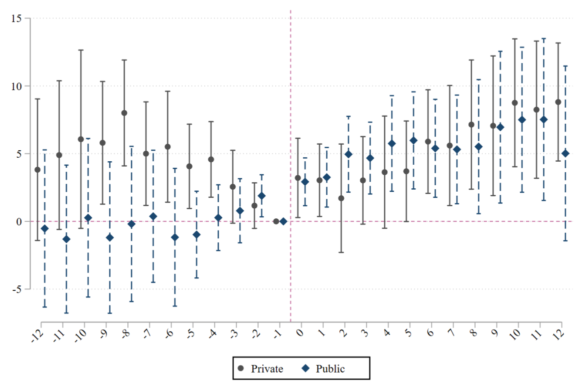
Figure A.3: Event Study for Graduates of Licensed Universities By Type of College



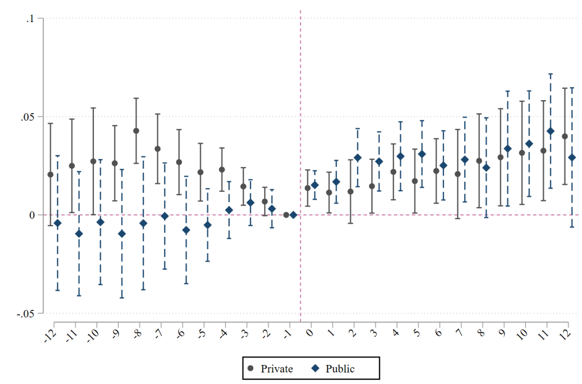
(a) Wages



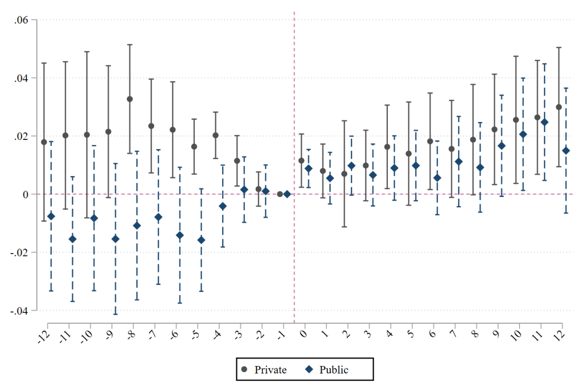
(b) Wages (Unconditional)



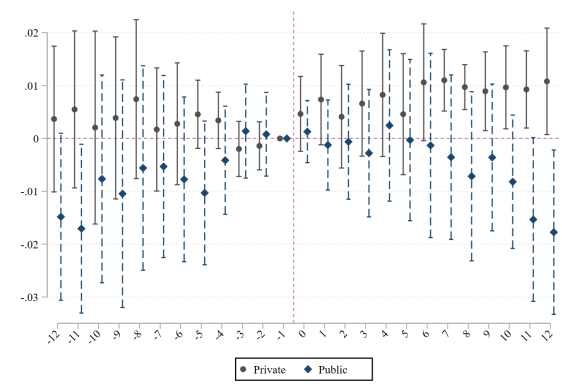
(c) Total Hours



(d) Employment

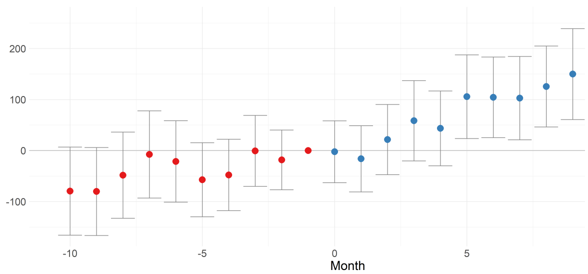


(e) Large Firm

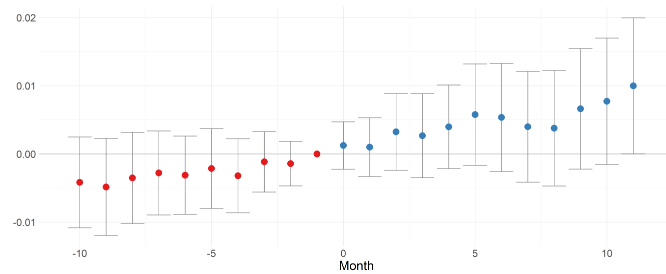


(f) Public Employer

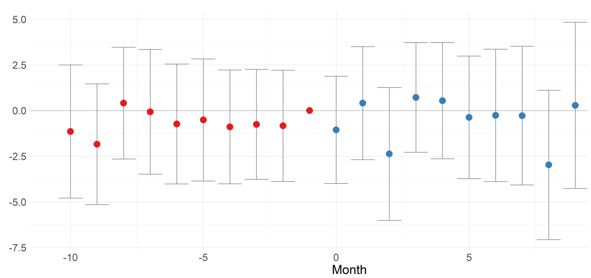
Figure A.4: Event Study for Graduates of Licensed Universities following Callaway and Sant'Anna (2021)



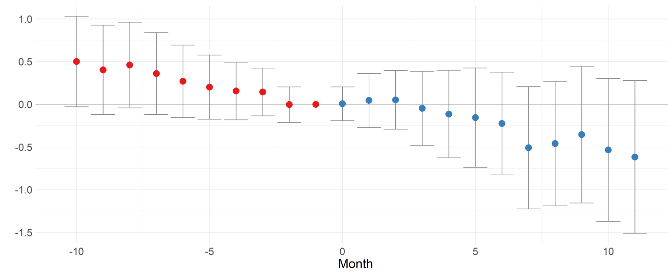
(a) Wages



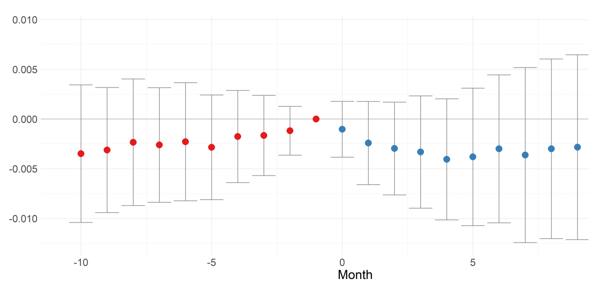
(b) Employment



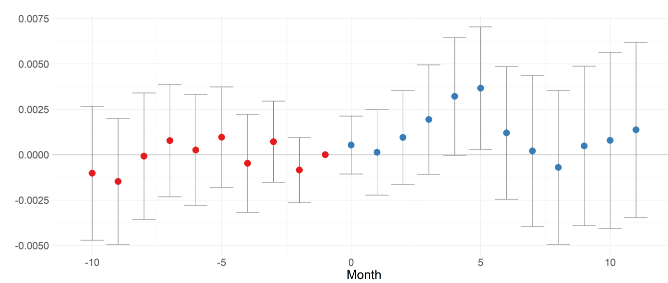
(c) Total Hours



(d) Tenure

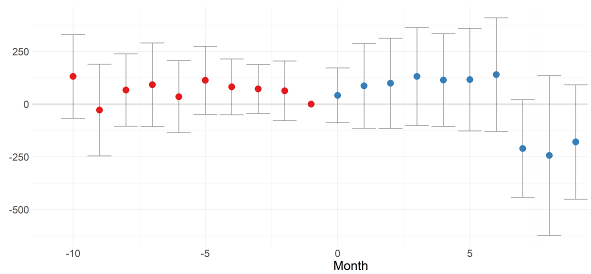


(e) Large Firm

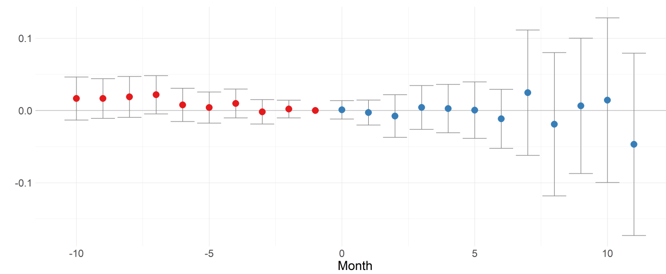


(f) Public Employer

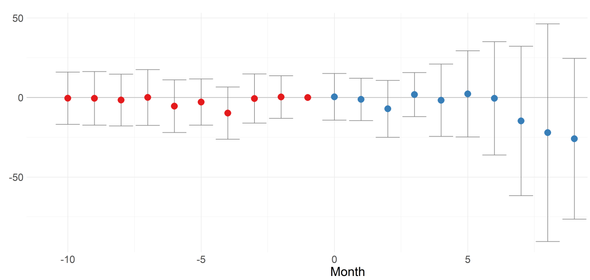
Figure A.5: Event Study for Graduates of Unlicensed Universities following Callaway and Sant'Anna (2021)



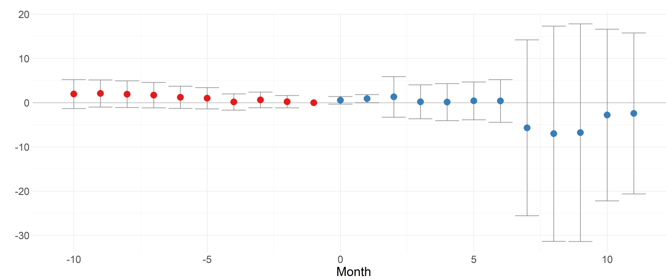
(a) Wages



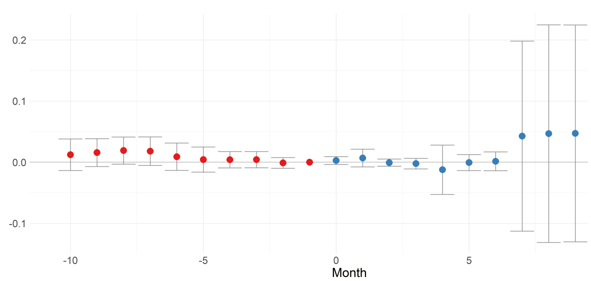
(b) Employment



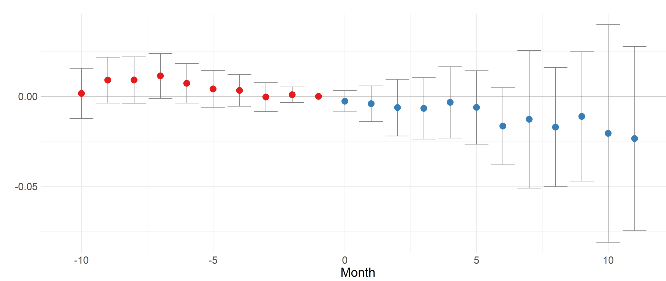
(c) Total Hours



(d) Tenure



(e) Large Firm



(f) Public Employer

A.2 Licensing Requirements

SUNEDU implemented a licensing process that demanded universities to comply with 8 main requirements. *All* requirements were mandatory and failing to comply in one was enough to get the licence denied.

The requirements were the following:

1. **Academic objectives.** Universities should have clear paths for students to obtain degrees and diplomas. They are also required to have an academic curriculum for each degree and academic plans to improve them.

2. **Academic plan.** Universities must offer degrees that respond to current labor market demands and that are relevant to the context. They should also offer degrees that have both human and economic resources which are sustainable over time.

3. **Infrastructure and equipment.** Universities must have a good standing infrastructure with rooms exclusively for institutional purposes and guaranteed utilities (water, electricity, internet access, and sanitation). They also should have relevant laboratories according to the degrees they offer.

4. **Research agenda.** Universities must also produce relevant research and have a registry of the research they produce. They were also required to have active research faculty and follow ethical research regulations.

5. **Faculty.** Universities should have no less than 25 percent of full-time faculty. Professors should have a transparent hiring process and have good academic records (an MA or a Ph.D. diploma).

6. **Complementary services.** Universities should also offer complementary services to students such as basic health services, a minimum of 3 sports groups,

cultural services, library access, social and psychological services.)

7. **Job placement mechanisms.** Universities should provide a job bank and search for strategic alliances with the public and private sectors. They should also have an office to follow up with graduates.

8. **Institutional transparency.** Universities must make publicly available the following: institutional goals, academic rules and calendars, admissions logistics (calendar and available seats), school fees, academic plans, faculty names, and faculty openings.

A.3 Predicting the Licensing Process

A.3.1 Predicting Licensing

In this section, we examine whether licensing decisions were predictable. To do so, we use data from the 2010 University Census (CENAUN), conducted several years before the onset of the higher education reform, combined with machine learning techniques to estimate which universities were more likely to obtain a license. The CENAUN provides comprehensive information on universities operating prior to 2010, including detailed data on infrastructure, faculty, administration, and student characteristics. Our analysis leverages approximately 100 variables from this dataset.

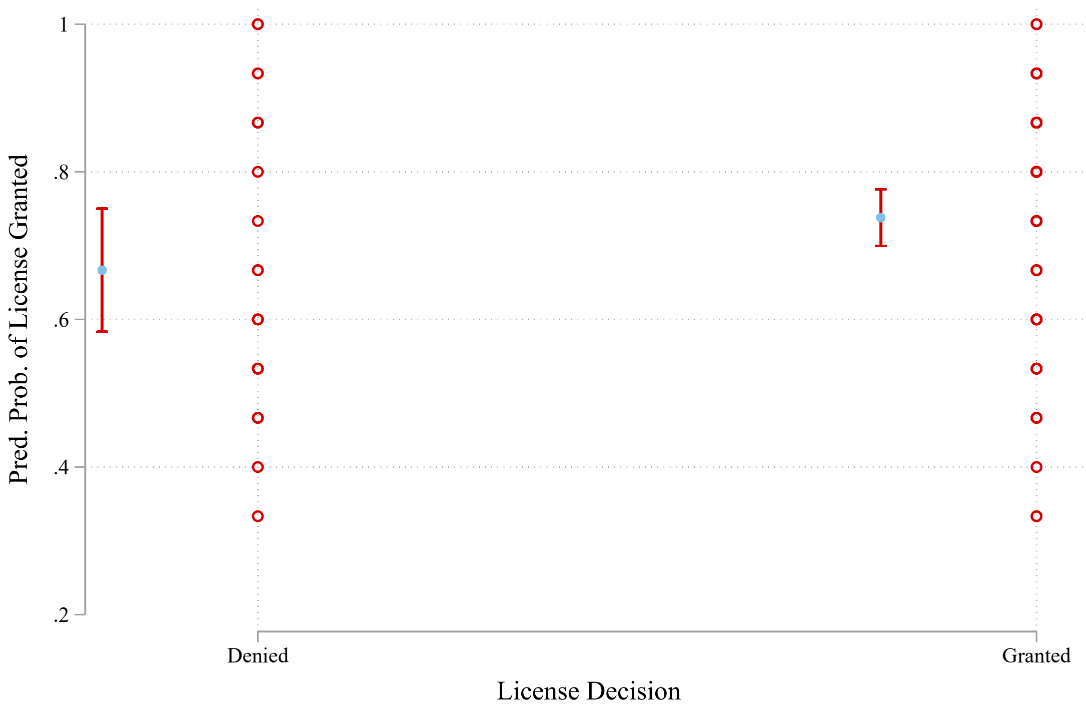
However, it is important to note that CENAUN is less likely to contain detailed information on institutions that were ultimately denied a license. As a result, our model may overestimate the probability of licensing, particularly for universities with incomplete or missing data.

First, we explore several machine learning methods to predict licensing outcomes, including Lasso regression, logistic regression, decision tree models, and the k-nearest neighbors (KNN) algorithm. Among these, the KNN algorithm achieved the highest predictive accuracy, with a test-set score of 0.92. We implemented k-fold cross-validation and selected the optimal parameter value of $k = 15$ using *GridSearch*.

Although the model performs well on average, it struggles to accurately predict outcomes for certain universities. Figure A.6 shows the distribution of predicted probabilities by licensing decision. On average, licensed institutions have higher predicted probabilities; however, there is substantial overlap between the two groups. This suggests that, while the model is relatively effective at identifying institutions that are highly likely to be licensed, particularly elite universities, as shown in Figure A.7, it is less reliable at distinguishing institutions that are ultimately denied a license.

We also benchmark the model's predictions against income-based measures—used as a proxy for college quality—and find similar patterns.

Figure A.6: Predicted Probability of Getting the License vs Real Licensing Decision

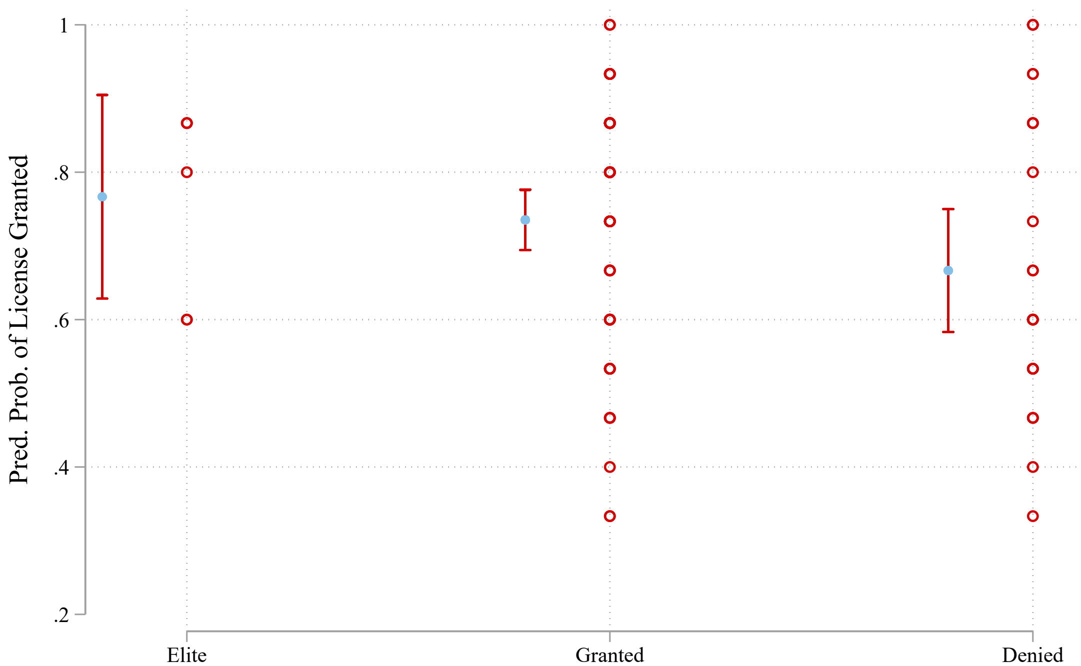


A.3.2 Predicting Timing if License

Second, we implement a survival analysis and time-to-event prediction using `PyTorch` to estimate the timing of licensing decisions. Specifically, we use a Cox-Time regression model—a continuous-time method that does not require discretizing the time scale. We also develop a simple neural network with two hidden layers, ReLU activations, batch normalization, and dropout.

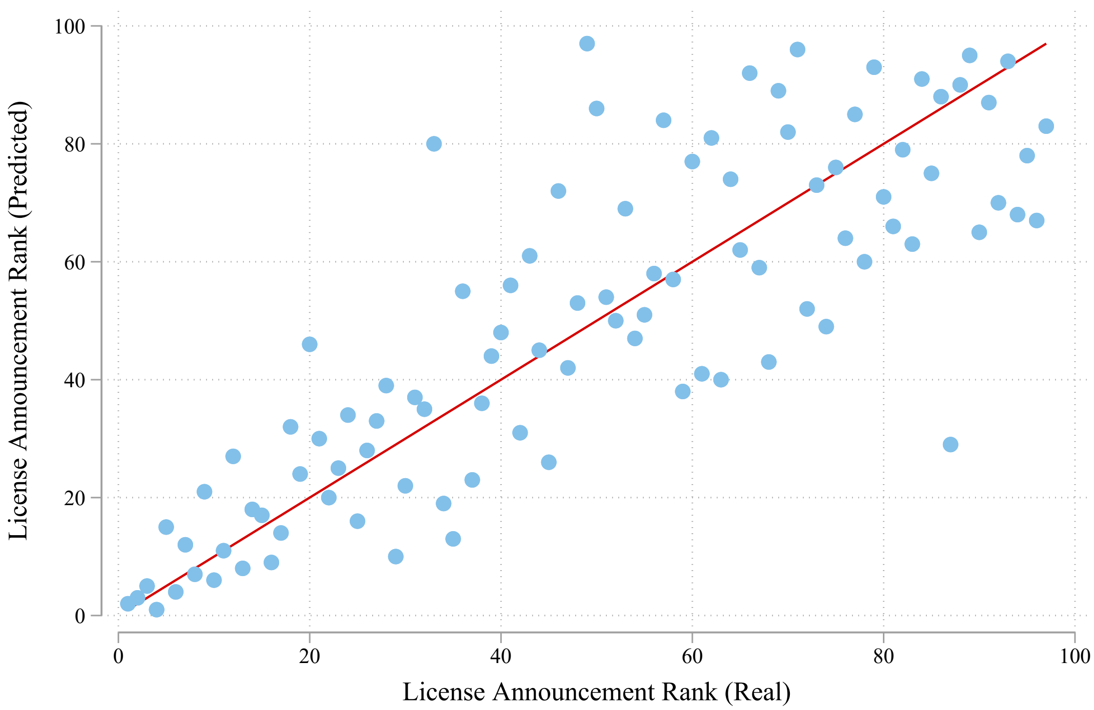
This framework allows us to estimate the probability that a university receives its licensing decision earlier than others. We use these probabilities to generate a predicted ranking of institutions based on the expected timing of their licensing decisions and

Figure A.7: Predicted Probability of Getting the License by Category



compare it with the actual order of announcements. As shown in Figure A.8, the correlation between the predicted and actual ranks is positive. However, the model is more accurate in predicting earlier licensing decisions than later ones. These early announcements largely correspond to elite universities, which also had high predicted licensing probabilities in the previous analysis.

Figure A.8: Predicted Probability of Getting the License First vs. Real



Note: this graph shows the correlation of the predicted rank of getting the licensed versus the actual rank.

A.4 ENAHO

Alternative Data Source: ENAHO Analysis

To assess the robustness of our main findings and address concerns about labor market informality, we complement our primary analysis with data from the Encuesta Nacional de Hogares (ENAHO), Peru's National Household Survey, covering the period 2014-2023. While ENAHO provides broader coverage of the labor market, including informal workers, it offers less granular temporal variation compared to our main dataset. Consequently, we exploit quarterly (rather than monthly) variation and employ the Callaway and Sant'Anna (2021) difference-in-differences estimator to construct event studies. Our analysis focuses on college graduates aged 45 or younger.

Results for Licensed Colleges

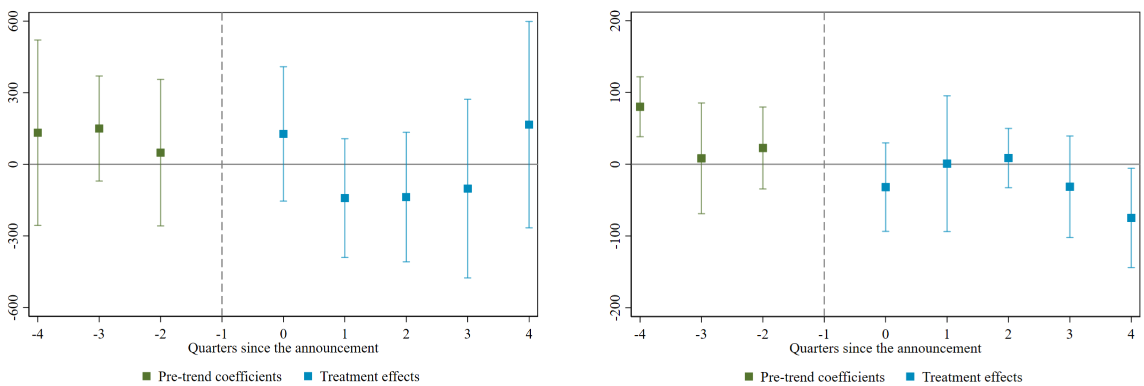
Figures A.9 through A.11 present event study estimates for graduates from licensed colleges using ENAHO data.

Wages. Panel (a) of Figure A.9 shows a modest but noisy increase in main job monthly wages in the fourth quarter following licensing, though the effect is imprecisely estimated with wide confidence intervals. Panel (b) reveals a small decrease in side job wages, again with substantial uncertainty around the point estimates.

Employment Outcomes. Figure A.10 indicates no discernible effects on overall employment (panel c), while panel (d) suggests a small, statistically insignificant decrease in self-employment rates following the licensing reform.

Informality. We examine two measures of labor market informality. Panel (e) of Figure A.11 employs a contract-based definition (workers without formal employment contracts), while panel (f) uses ENAHO's standard informality measure. Neither specification reveals statistically significant effects of licensing on informality rates among graduates from licensed institutions.

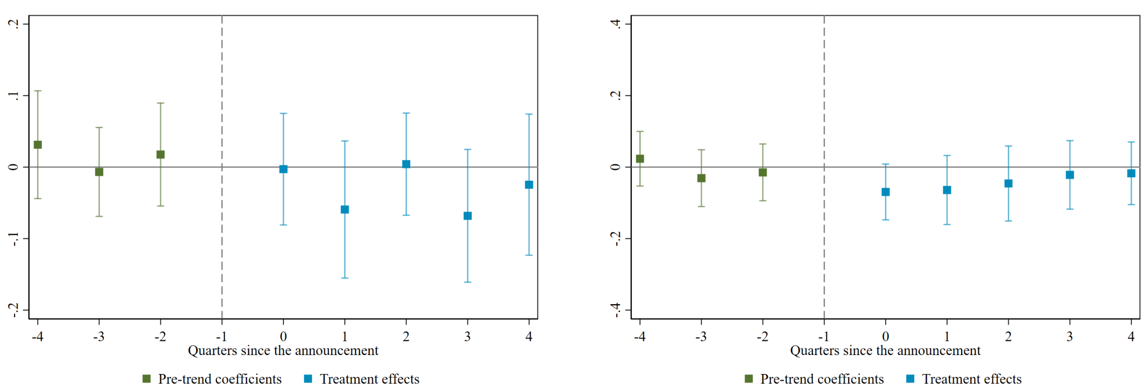
Figure A.9: Event Study Estimates: Wage Effects for Licensed Colleges (ENAHO)



(a) Wages (main job, monthly PEN)

(b) Wages (side job, monthly PEN)

Figure A.10: Event Study Estimates: Employment Effects for Licensed Colleges (ENAHO)



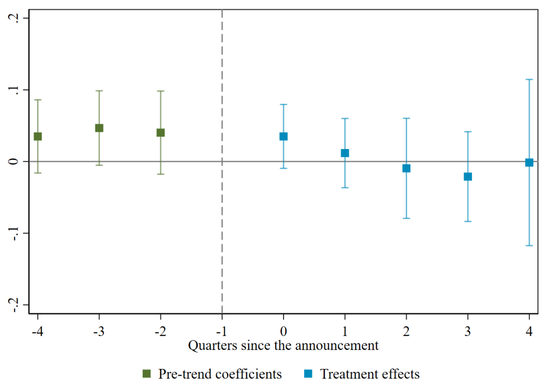
(a) Employment

(b) Self-Employment

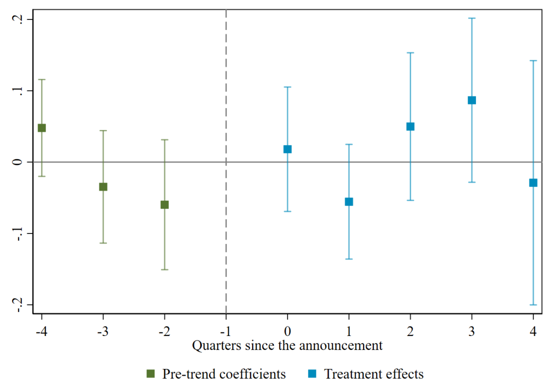
Results for Non-Licensed Colleges

Figures A.12 through A.14 present analogous event study estimates for graduates from non-licensed colleges.

Figure A.11: Event Study Estimates: Informality Effects for Licensed Colleges (ENAHO)



(a) Informality (no contract)



(b) Informality (ENAHO)

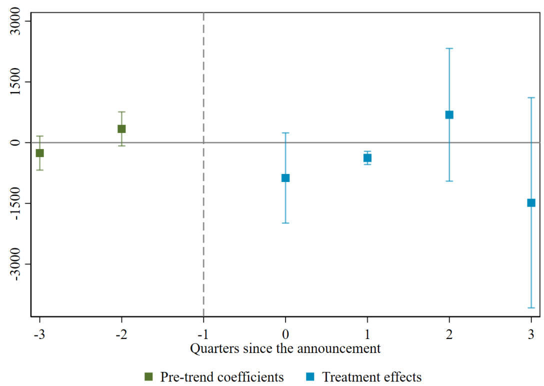
Wages. Panel (a) of Figure A.12 shows small declines in main job wages for graduates from non-licensed institutions in the post-reform period. Conversely, panel (b) indicates an increase in side job wages in the third quarter, suggesting potential labor market adjustments through secondary employment.

Employment Outcomes. Figure A.13 reveals no statistically significant effects on overall employment (panel c), while panel (d) shows a small decrease in self-employment rates, though the magnitude is modest.

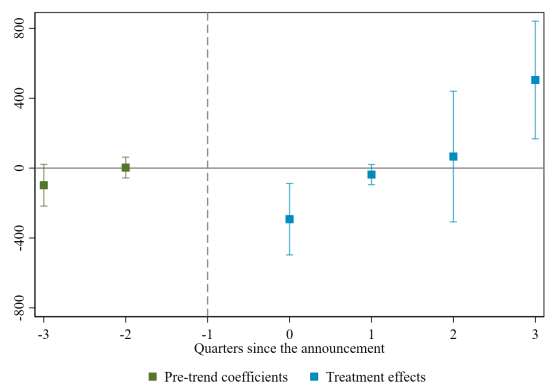
Informality. Unlike the licensed college sample, graduates from non-licensed institutions exhibit noisy increases in the likelihood of informal employment across both measures (Figure A.14, panels e and f). However, these effects are estimated with considerable uncertainty, and the confidence intervals include zero for most periods.

The ENAHO-based results provide suggestive evidence consistent with our main findings, though the estimates are generally noisier due to reduced sample sizes and coarser temporal variation. For licensed colleges, we observe patterns suggesting modest wage gains and no systematic changes in informality. For non-licensed colleges, the

Figure A.12: Event Study Estimates: Wage Effects for Non-Licensed Colleges (ENAHO)

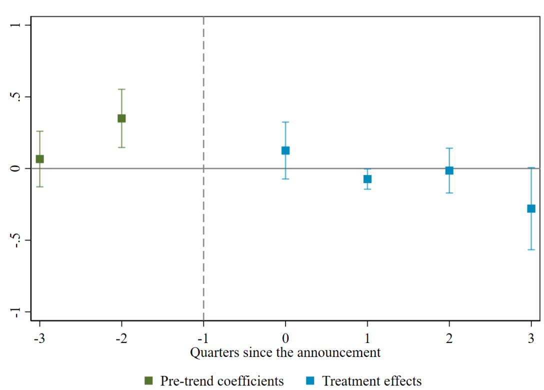


(a) Wages (main job, monthly PEN)

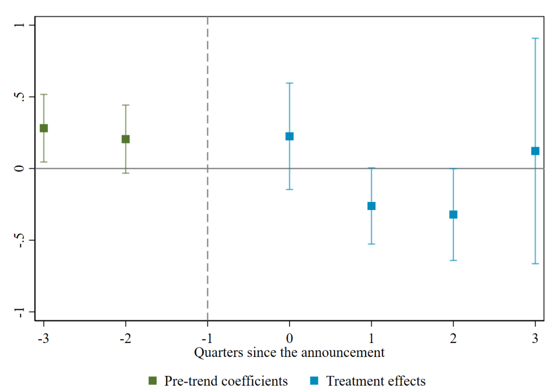


(b) Wages (side job, monthly PEN)

Figure A.13: Event Study Estimates: Employment Effects for Non-Licensed Colleges (ENAHO)

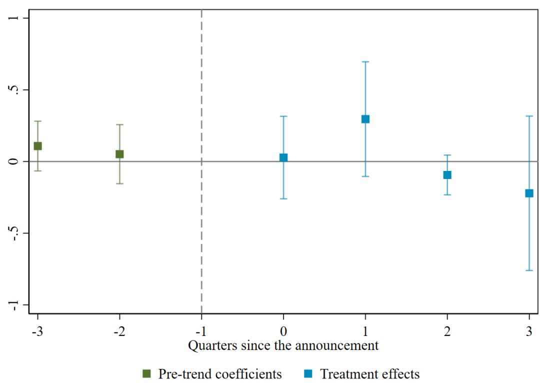


(a) Employment

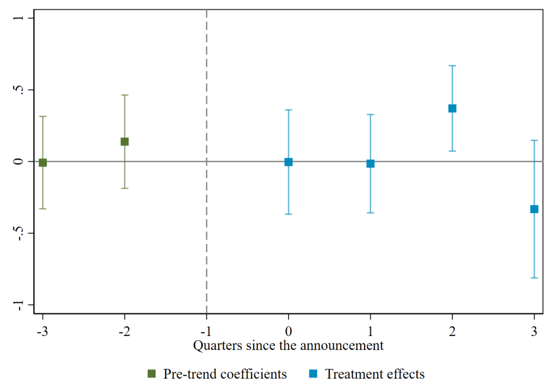


(b) Self-Employment

Figure A.14: Event Study Estimates: Informality Effects for Non-Licensed Colleges (ENAHO)



(a) Informality (no contract)



(b) Informality (ENAHO)

results hint at potential adverse effects on main job wages and increases in informal employment, though statistical power limitations prevent definitive conclusions. These patterns align qualitatively with our primary analysis while highlighting the importance of data granularity for detecting labor market effects of education policy reforms.